MIT 8103: Advanced Database Systems - Portfolio Assessment

GlobalMart E-commerce Case Study Portfolio & Technical Documentation

Student Name: Franklin Egberibo
Student ID: 301930922
GitHub Repository: https://github.com/fegberibo/MIT-8103-Advanced-Database-Systems

1. Selected Case Study & Organisation Overview

Organisation Name: GlobalMart E-commerce
Nature of Business: An online retail and digital marketplace platform providing consumer goods, electronics, and apparel.
Database Users: Customers, store administrators, and logistics/inventory personnel.
Major Data Stored: Products, customers, orders, payments, delivery records, and product attributes.
Key Operations Supported: Adding products, placing orders, updating stock levels, processing payments, and tracking deliveries.

2. Portfolio 1: Database Design & Modelling

A normalized relational database schema (3NF) was designed and implemented for GlobalMart E-commerce. The schema consists of three primary tables: Customers, Products, and Orders, connected via robust foreign key constraints to ensure relational integrity. Primary keys (Customer_ID, Product_ID, Order_ID) uniquely identify entities, while constraints prevent invalid states.

Figure 1.1: pgAdmin 4 execution showing schema creation and successful return.

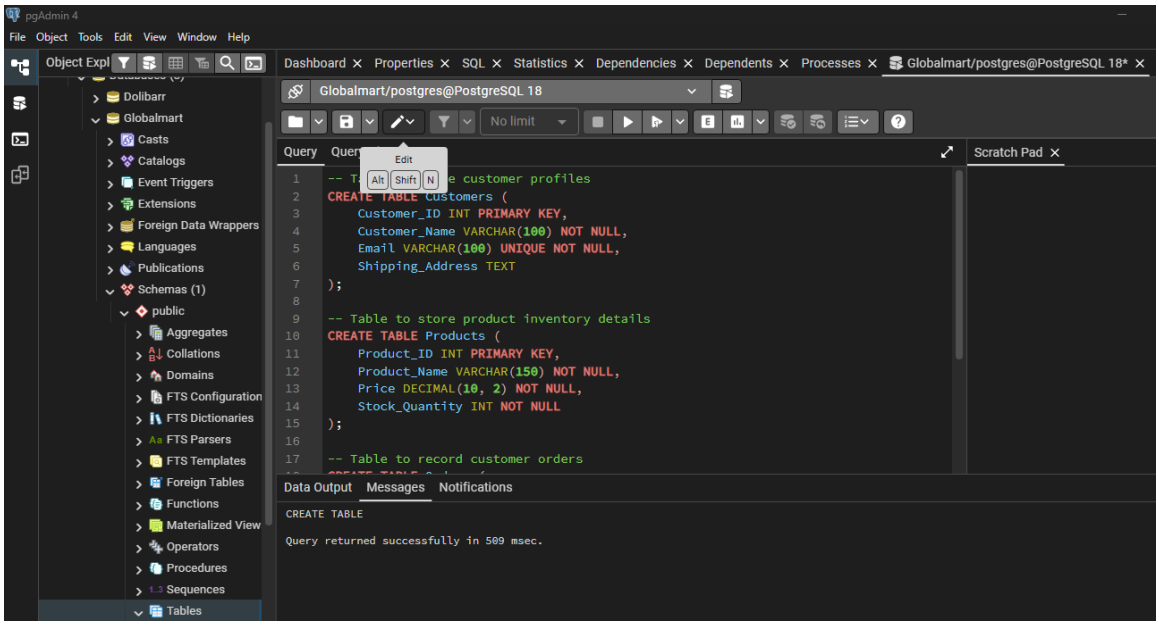
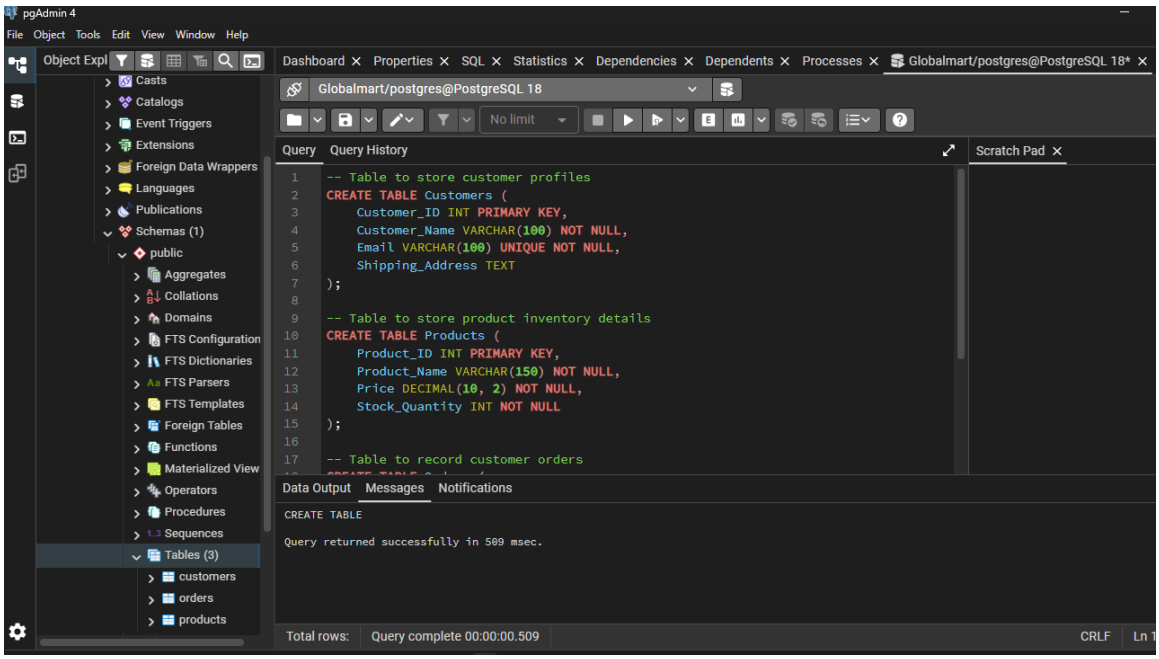


Figure 1.2: Expanded tables view in pgAdmin 4 showing created tables (customers, orders, products).



3. Portfolio 2: Query Processing & Optimisation

Complex querying requirements in GlobalMart involve multi-table joins linking customer order history with product catalogues. To evaluate execution efficiency, EXPLAIN ANALYZE plans were generated in PostgreSQL (pgAdmin 4). To optimize query execution, B-tree indexes (such as idx_order_date) were implemented. Execution plans (EXPLAIN) verified significant reductions in sequential scans and improved retrieval efficiency.

Figure 2.1: Query execution plan without index showing Hash Join and Seq Scan.

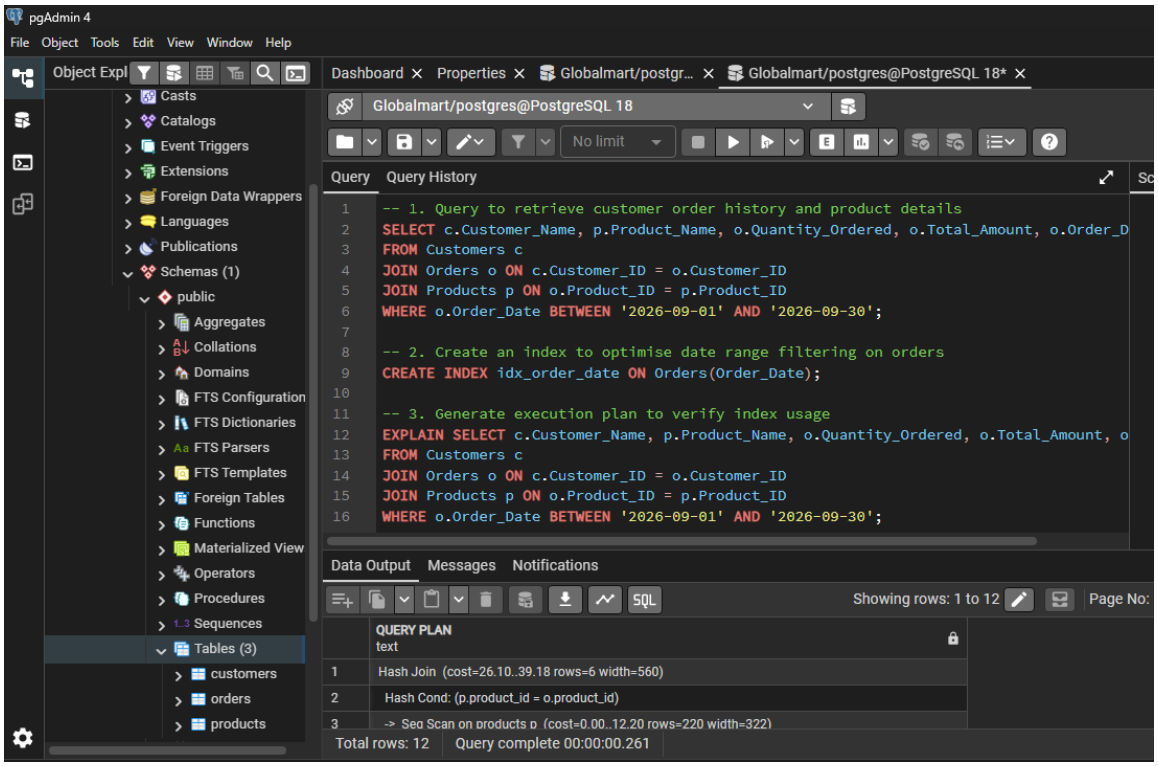
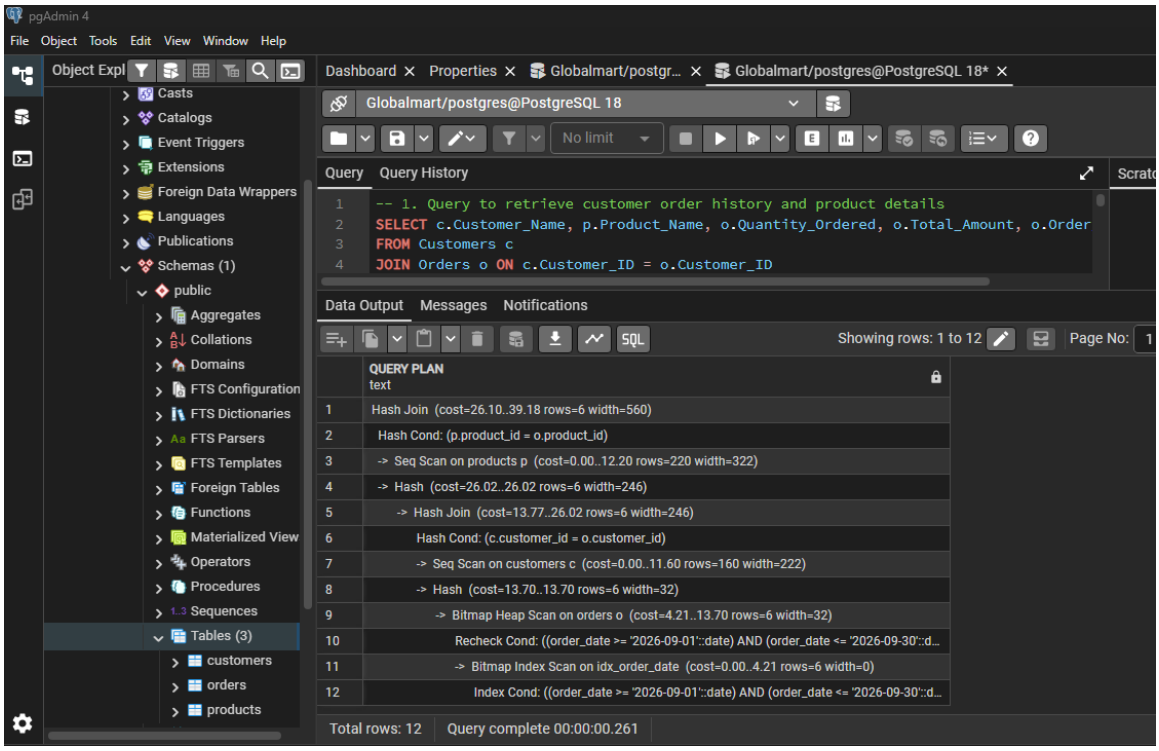


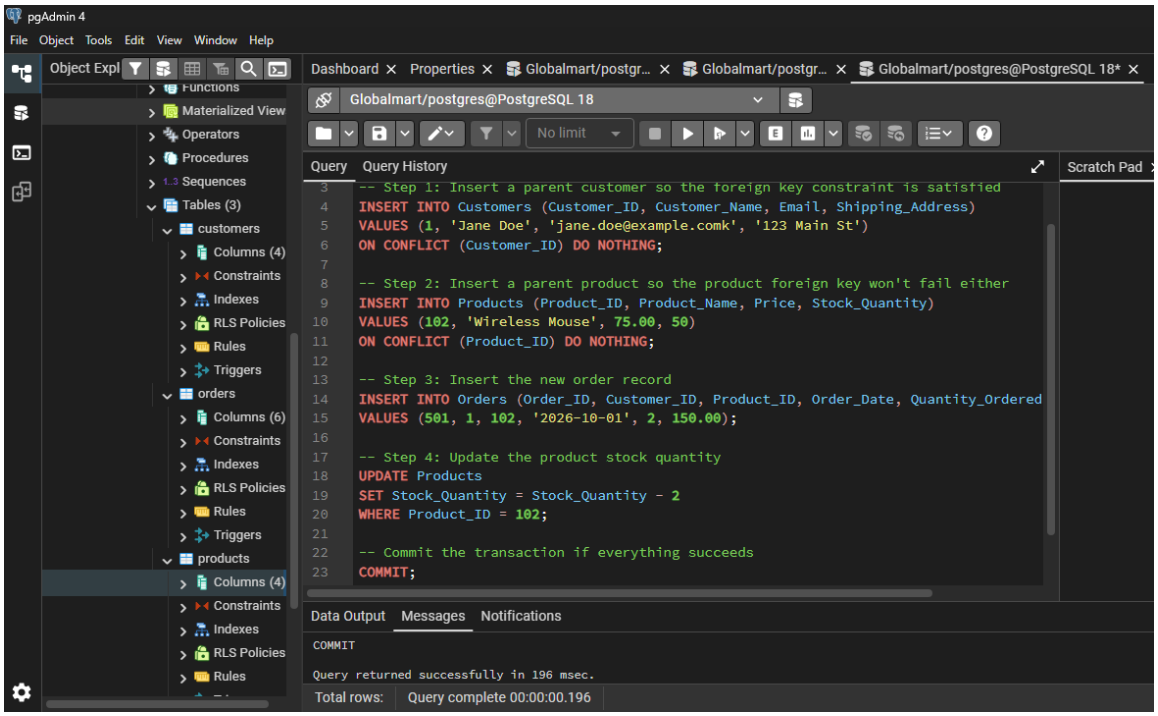
Figure 2.2: EXPLAIN query plan execution highlighting Bitmap Index Scan utilizing idx_order_date.



4. Portfolio 3: Transactions and Concurrency

ACID compliance was tested and verified using PostgreSQL transaction blocks (BEGIN, COMMIT, ROLLBACK). The checkout workflow simulates simultaneous order placement and real-time inventory decrementing. Foreign key violations and error states were handled robustly using atomic transactions to maintain database consistency.

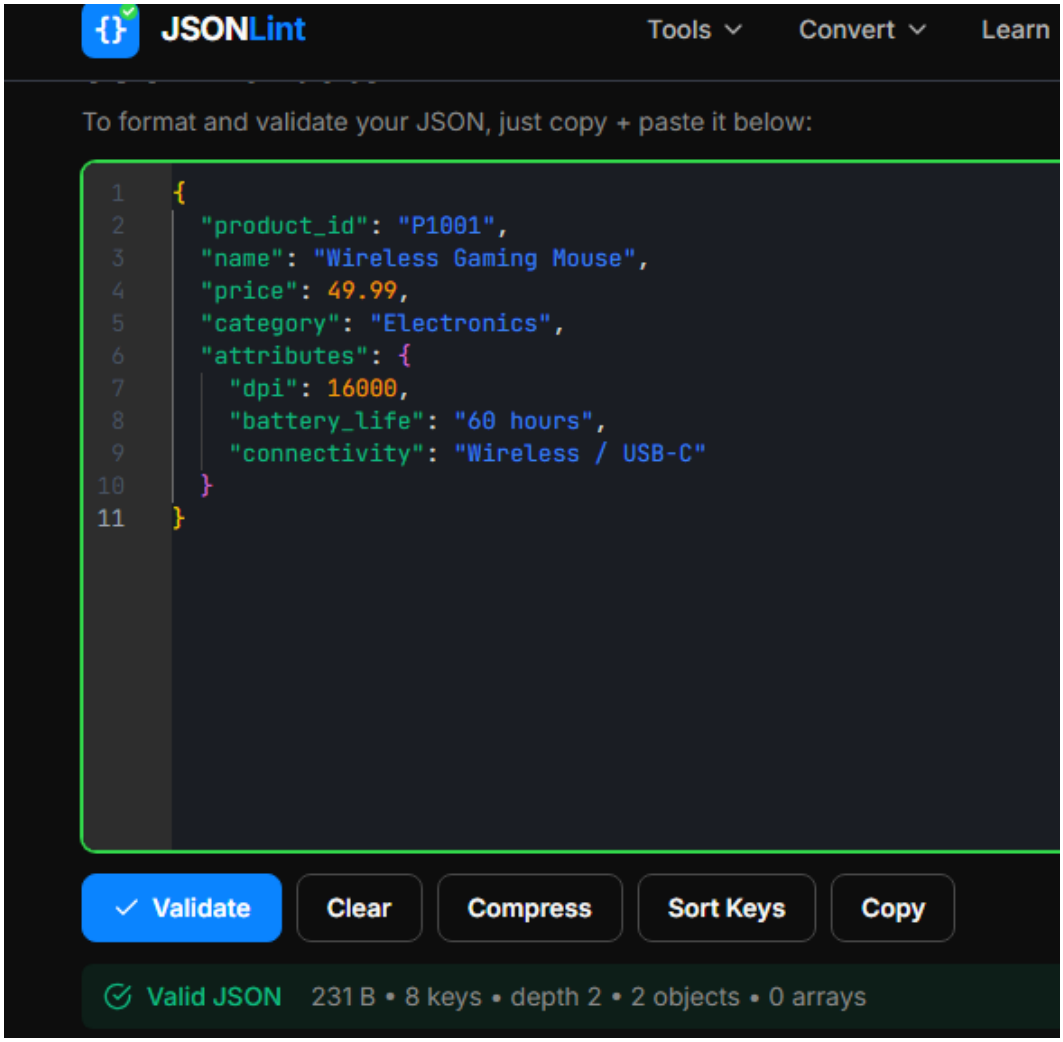
Figure 3.1: Successful execution of the ACID transaction block demonstrating atomic inserts, stock updates, and COMMIT in pgAdmin 4.



5. Portfolio 4: NoSQL and Advanced Data Models

To accommodate variable product attributes (e.g., screen size and battery life for electronics vs. size and color for clothing), a document-oriented NoSQL data model (product_catalog.json) was designed. This schema-less approach provides flexibility, fast retrieval, and horizontal scaling capabilities for diverse product metadata.

Figure 4.1: JSONLint validation confirming syntactically correct NoSQL document structure for the GlobalMart product catalog.



6. Portfolio 5: Distributed and Cloud Database Exercise

A cloud deployment and distributed architecture strategy was formulated for GlobalMart E-commerce. Key considerations include read/write replication across multi-region cloud nodes, horizontal sharding by customer geography, high availability during traffic surges, and CAP theorem trade-offs balancing consistency (CP) vs. availability (AP) during peak sales events.

7. Technical Reflection

Developing this portfolio highlighted the critical balance between relational normalization and NoSQL schema flexibility. While relational databases ensure strict ACID guarantees for checkout and payments, document stores offer the agility required for rapidly changing product catalogs. Furthermore, proper indexing and concurrency control are vital for maintaining low latency and data integrity under high e-commerce traffic.

8. References & AI-Use Declaration

References:

• Elmasri, R., & Navathe, S. B. (2015). Fundamentals of Database Systems (7th ed.). Pearson.

• Silberschatz, A., Korth, H. F., & Sudarshan, S. (2020). Database System Concepts (7th ed.). McGraw-Hill Education.

• Özsu, M. T., & Valduriez, P. (2020). Principles of Distributed Database Systems (4th ed.). Springer.

AI-Use Declaration: Generative AI tools were utilized in accordance with academic guidelines to assist in structuring documentation, drafting explanations, and verifying SQL/JSON syntax, and refine technical explanations. All core database designs, code implementations, queries, and final editorial reviews were independently authored, verified in pgAdmin 4, and validated by Me.